

Tribhuvan University
Faculty of Humanities & Social Sciences
OFFICE OF THE DEAN

2022

Bachelor in Computer Applications

Course Title: **Mobile Programming**

Code No: **CACS351** Semester: **VI**

Full Marks: **60** Pass Marks: **24** Time: **3 hours**

Candidates are required to answer the questions in their own words as far as practicable.

Note: Group A (Multiple Choice Questions) is not available from the source. The paper begins with Group B.

Group B

2. What is a smart mobile device? Explain different platforms used in mobile devices.
3. What is a surface manager library? How DVM is different from ART?
4. What are important attributes used in Absolute Layout? Develop a simple calculator application with two input fields for inputting numbers and two buttons for performing multiplication and division and display the result in TextView using Absolute Layout.
5. What is the difference between activity and fragment? Develop an android application to create a user interface for input length, width, and height of a Rectangular box and Button for Volume in one fragment when the button is clicked then these input data are passed into the second fragment and calculate the volume of the Rectangular box and display the result in TextView.
6. What is Toast? Develop an android application to create an alert dialog to input any number and then calculate the factorial of the number and display the result in Toast.
7. What are different data types supported by SQLite? How to decode JSON data in an android application? Explain with a suitable example.
8. What do you mean by View hierarchy in IOS programming? Write a swift program to find the sum of all elements of a 1-D array.

Group C

Attempt any TWO questions[2x10=20]

9. What is the difference between ArrayAdapter and CursorAdapter? Develop an android application to display the application name with logo (i.e., YouTube, Facebook, Twitter, Tiktok, Snapchat, Telegram etc.) in Grid View and display their site in another activity when a Grid View item is clicked.
10. Why we use API in android? Provided that SQLite database named "Hospital" with table named Doctor with the following columns (Did as Integer, Dname as Text, and Specialization as Text and Experience as Real). Develop an android application to connect to the database and insert records of Doctor and display the

Doctor information whose experience is less than 5.5 years.

- 11.** What does android APK file contain? Develop an android application with a context menu having menu items "Red", "Yellow", "Green", "Black" and "Blue" etc. and change the background color of the layout when the user selects a menu item accordingly.